① The Space of Complementary Techique 127

① What is the "Space of Complementary Techniques"?

→ Think of A/B testing as your main diagnostic tool - it tells you if a change moves the needle. But it can't tell you why, for whom, or under what conditions. The "space" of complementary techniques is simply the set of other methods you bring in to fill those gaps:

- Logs-Based Analysis (quantitative, high scale)
- Human Evaluation (qualitative, expert judgement)
- User-Experience Research (contextual, interviews observation)
- Focus Groups (guided group discussion)
- Surveys (structured self-report)
- External Data (look at outliers or power users).

Mathematically, you can think of each technique as providing a vector of insight. Some vectors are more aligned (similar) and some more orthogonal (complementary). By combining them, you span a richer "insight space" than with A/B testing alone.

② Real-World Analogy: Building a Balanced Meal.

- A/B Test = Main protein (tells you your body is getting fuel.
- Logs Analysis = Calories (gives volume — how much you consumed).

- Human Eval & UBR = Vegetables (add nuance — micronutrients and flavors).
- Surveys & Focus Groups = Spices & herbs (gather subjective taste preferences).
- External Data = Dessert (high-impact, but only for certain palates)

Just as a balanced meal keeps you healthy, a balanced mix of methods gives you robust, multi-angle insights.

(3) Underlying Math: Vector Spaces & Orthogonality.

What Does "space of complementary Techniques" Mean?

→ In A/B testing, you mostly get what happened (eg: "conversion rate increased by 2%")
But that's not enough when you need:
- Why did it happen?
- Who did it affect most?
- Was the experience better or worse?
- Any side effects on long-term risks?

So, we use complementary techniques. Each technique one adds a different type of insight.

Simple Analogy: People Describing a Room.

Imagine 3 people enter a room and report on it:
- Engineer: "The room is 4m x 5m, light intensity is 300 Lux" (Quantitative)
- Artist: "It feels warm and cozy" (Qualitative).
- Psychologist: "People in here tend to relax" (Behavioral Insight)

Each person gives a different dimension of the truth. This is exactly like compare meaning technique in experimentation.

How Math Connects: Vectors and Dimensions

In math, a vector is just a list of values that point in a direction in space.

Let's map each technique as a vector in an "insight space":

- logs-based analysis → captures numbers, metrics
- Survey → captures opinions
- focus group → captures emotions, reactions
- UGK → captures context, usability pain

Each method adds a new dimensions.

Visual: 2D Analogy

Let's say:

- Logs = Vector along x-axis → [1, 0]
- Surveys = Vector along y-axis → [0, 1]

These two are orthogonal (perpendicular) i.e. they don't overlap — they bring completely different information.

Now imagine:

- A third method like focus group → lies between x and y → [0.7, 0.7]

→ This overlaps partially with Logs and Surveys.

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Math Explanation: Dot Product

1. Vectors in $\mathbb{R}^n$ (n-dimensional real space)

Each vector represents a technique (e.g. logs, surveys, focus groups) with components in different informational dimensions.

Example in $\mathbb{R}^2$:

$$a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

2. Dot Product Definition

The dot product (inner product) of two vectors a and b is defined as:

$$a \cdot b = \sum_{i=1}^{n} a_i b_i$$

In $\mathbb{R}^2$:

$$a \cdot b = a_1 b_1 + a_2 b_2$$

3. Geometric interpretation

The dot product can also be defined using the angle θ between the two vectors:

$$a \cdot b = \|a\| \, \|b\| \cos\theta$$

where

- $\|a\| = \sqrt{a_1^2 + a_2^2}$ is the Euclidean norm (length)
- $\cos\theta$ = measure of alignment.

4. Orthogonality

if $\theta = 90°$, then $\cos\theta = 0$, so:

$$a \cdot b = 0$$

This means vectors are orthogonal ⇒ techniques are completely independent.

5. Normalization and Cosine Similarity.

To compare direction only (not magnitude), normalize both vectors:

$$\hat{a} = \frac{a}{\|a\|}, \quad \hat{b} = \frac{b}{\|b\|}$$

Then

$$\hat{a} \cdot \hat{b} = \cos\theta \quad (\vec{a} \cdot \vec{b} = \cos\theta)$$

This is called cosine similarity, values range

- 1 : same direction
- 0 : orthogonal
- -1 : opposite direction.

6. Concrete Example in $\mathbb{R}^2$

Let's define:

Logs : $l = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, Surveys : $s = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, Focus Group : $f^2 = \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$

Compute dot products:

Logs · Survey $= 1 \cdot 0 + 0 \cdot 1 = 0$ (orthogonal)

Logs · F.G $= 1 \cdot 0.7 + 0 \cdot 0.7 = 0.7$

Survey · F.G $= 0 \cdot 0.7 + 1 \cdot 0.7 = 0.7$

These indicates that focus group overlaps partially with both logs and surveys (cosine = 0.7), while logs and surveys are completely independent.

7. Matrix View of Technique Space

Let all technique be rows of a matrix

$$T = \begin{bmatrix} 1 & 0 \\ (\text{Logs}) & \\ 0 & 1 \\ (\text{Surveys}) & \\ 0.7 & 0.7 (\text{Focus}) \end{bmatrix}$$

Let is construct the Gram matrix

$$G = TT^T$$

Each entry G_{ij} = dot product between technique i and j.

Why this Matters in Real Experiments

If you only use logs, you get only the what (e.g. clicks, conversions).

But you need complementary dimension like:

- Human eval: Was the new feature ugly?
- UER: Was the button hidden on mobile?
- Focus group: Did users misunderstand the new flow?

Using complementary techniques is like adding more dimension to flat pictures to build a 3D understanding.

Final Summary

Concept	Meaning
Technique	A method for analyzing impact (e.g. logs, survey)
Vector	Mathematical way to represent info direction / dimension.
Dot Product	How similar two vectors (technique) are.
Orthogonal (dot = 0)	Method give completely different complementary insights.
Aligned (dot ≠ 1)	Methods overlap, give similar insight.
Insight Space	Multi-dimension view build from multiple technique.

(4) Overlaps with Other Topic

- Meta-Analysis (Unit 8): When you aggregate multiple A/B test over time, you're essentially combining vectors from successive experiments - just like combining complementary technique.
- Institutional Memory (Unit 6): Storing past logs - analysis results, human-eval reports, and survey data is akin to keeping basis vectors in your insight space so that new signals can be projected against them.

(2) Logs-Based Analysis

(1) What is logs-Based Analysis

→ A logs-based analysis mines the exhaustive event logs your product already records - page views, clicks, scroll depth, API calls, latency, errors - to answer questions that an A/B test alone can't: who did what, in what order, how long they stayed, and how side-effect propogate.

(2) Why it matters

- Speed & scale: You can query billions of rows minutes after launch - no extra instrumentation if logging is standardized.
- Granularity: Unlike the single binary "converted/didn't" in an experiment, logs expose real user journeys and rare edge cases.
- Diagnosis: If an A/B test's headline metric moved, logs tell you which step in the funnel broke or improved.
- No-test insights: You can spot regressions or opportunities even when no formal experiment is running.

(3) Real-world analogy (and the math behind it)

Imagine a traffic camera grid in a city

Piece	Analogy	Math view.
Each camera frame	One log line	A point in a high-dimensional space of attributes
Vehicle trajectory	User session	A sequence $(x_1, x_2, \ldots x_T)$
Counting cars at an intersection	Computing a metric (e.g. click-through rate)	Estimator $\hat{p} = \frac{\sum \text{clicks}}{\sum \text{views}}$.
Comparing two days's traffic	Pre/post or A/B diff	Difference-in-means, variance from CLT.

Mathematically, a log is a stochastic process $\{X_t\}_{t=1}^{T}$. Summaries like totals, means, funnels, or retention curves are statistics — function $g(\{X_t\})$ whose sampling distributions you can approximate with the Central Limit Theorem or bootstrap.

(4) Core workflow

Row	User	Time	Event
1	A	10	view
2	A	12	click
3	A	60	view
4	A	95	click
5	B	20	view
6	B	55	view
7	C	18	view
8	C	19	click
9	C	75	view

Assume session timeout $T = 30$ minutes.

Step 1 : sessionize

Let the raw log be

$$L = \{(u_m, t_m, a_m)\}_{m=1}^{M}$$

where u_m is user-ID, $t_m \in \mathbb{R}$ the event time, and $a_m \in \{\text{view}, \text{click}, \ldots\}$ the event type.

Choose a session-timeout parameter $T > 0$ (e.g. 30 minutes). For each user u, sort events by time and put a session break whenever the gap exceed T:

$$t_{m+1} - t_m > T \Rightarrow \text{new session}$$

This yields sessions

$$S_{u,k} = \{(u_m, t_m, a_m) : u_m = u;\ t_m \in T_{u,k}\},$$

where the intervals $T_{u,k}$ are maximal contiguous blocks separated by gaps $> T$.

Examples:

- User A
  - session 1 : [10 (view), 12 (click)]
  - session 2 : [60 (view), 95 (click)] $\leftarrow 60 - 12 > 30$ → new session
- User B : [20, 55] → $55 - 20 = 35$ → new session
  - session 1 : [20 (view)]
  - session 2 : [55 (view)]

- User C
  - session 1 : [18 (view), 19 (click)]
  - session 2 : [75 (view)]

Step 2: Define session-level Metric.

For click-through rate (CTR) at the session level

$$g(s_{u,k}) = \frac{\sum (u_m, t_m, a_m) \in s_{u,k} * \{a_m = \text{click}\}}{\sum (u_m, t_m, a_m) \in s_{u,k} 1 \{a_m = \text{view}\}}$$

with the convention 0/0 := 0

You may similarly define, e.g. time-to-first-byte.

$$h(s_{u,k}) = \min \{ t_m - t_{req} : a_m = resp \}.$$

Session	Views	clicks	CTR $g(s_{u,k})$
A session 1	1	1	1.0
A session 2	1	1	1.0
B session 1	1	0	0.0
B session 2	1	0	0.0
C session 1	1	1	1.0
C session 2	1	0	0.0

Step 3: Stratify

Stratify means:

> Grouping data into separate subgroups based on some important property – like new vs returning users.

Let's define an indicator variable

$$Z_u = \begin{cases} 1 & \text{if user } u \text{ is new} \\ 0 & \text{if user } u \text{ is returning.} \end{cases}$$

Why Do we Stratify?

In A/B testing:

- Some types of users behaves differently (e.g. new users might click more)
- If more new users randomly land in Group B, it might look better, even if the experiment had no real effect.
- To fix this, we split sessions by user type, analyse separately, then combine results.

** Steps to Determine Z (indicator) **

For each user.

1. Check the earliest date/time they ever appeared in your logs.
2. If they joined during the experiment, Label them:
   - Z = 1 → New user
3. If they used the app before the experiment started, label them
   - Z = 0 → Returning user.

* We only assume: *

- user A = Returning → $z_A = 0$
- user B = New → $z_B = 1$
- user C = New → $z_C = 1$

Stratified sets:

- S_{new}: Sessions from B & C
- S_{ret}: session from A.

Step4: Compare Cohorts (A/B)

Assume each session belongs to control (A) or treatment (B) via

$$C_{u,k} \in \{A, B\}$$

Define the cohort-level CTR means

$$\hat{\mu}_A = \frac{1}{n_A} \sum_{C_{u,k}=A} g(S_{u,k}), \quad \hat{\mu}_B = \frac{1}{n_B} \sum_{C_{u,k}=B} g(S_{u,k})$$

with n_A, n_B the cohort sample sizes.

With session independence, the two-sample z-statistic is

$$Z = \frac{\hat{\mu}_B - \hat{\mu}_A}{\sqrt{\frac{\hat{\sigma}_A^2}{n_A} + \frac{\hat{\sigma}_B^2}{n_B}}}, \quad \hat{\sigma}_A^2 = \frac{1}{n_A - 1} \sum_{C_{u,k}=A} (g(S_{u,k}) - \hat{\mu}_A)^2,$$

and analogously for $\hat{\sigma}_B^2$

Reject $H_0: \mu_A = \mu_B$ at level 0.05 if $|Z| > 1.96$.

* How are Users Assigned to Control / Treatment?

In real A/B testing:

- Users are randomly assigned to group using a hash function (often on User ID or Cookie)
- This ensures that the experiment is fair and unbiased.

Let's assume the experiment randomly assigned:

User	Group Assigned
A	Treatment (B)
B	Control (A)
C	Treatment (B)

** This is not based on click/view behavior, but done before the experiment starts - it's like flipping a fair coin for each user. **

Then:

- Group A sessions: B's: CTRs = [0.0, 0.0]

$$\bar{\mu}_A = \frac{0+0}{2} = 0.0$$

- Group B sessions: A + C's: CTRs = [1.0, 1.0, 1.0, 0.0]

$$\bar{\mu}_B = \frac{1+1+1+0}{4} = 0.75$$

Step 4.1: Compute variance

$$\hat{\sigma}_A^2 = \frac{(0.0)^2 + (0.0)^2}{2-1} = 0.$$

$$\hat{\sigma}_B^2 = \frac{(1-0.75)^2 + (1-0.75)^2 + (1-0.75)^2 + (0-0.75)^2}{4-1}$$

$$= \frac{3(0.25^2) + 0.75^2}{3} = 0.25.$$

Step 4.2 : Compute Compute Z-statistic.

$$Z = \frac{\hat{\mu}_B - \hat{\mu}_A}{\sqrt{\frac{\hat{\sigma}_B^2}{n_B} + \frac{\hat{\sigma}_A^2}{n_A}}} = \frac{0.75 - 0}{\sqrt{\frac{0.25}{4} + 0}} = \frac{0.75}{\sqrt{0.0625}}$$

$$= \frac{0.75}{0.25} = 3.0$$

Step 4.3 Conclusion.

Since $|Z| = 3.0 > 1.96$, we reject null hypothesis at 95% confidence level.

☑ CTR of treatment (Group B) is significantly higher than control (Group A).

In statistics

- if $Z > 1.96$ → the result is statistically significant at 95% confidence
- if $Z < 1.96$ → difference could just be random chance

Step 5 : Anomaly Detection (EWMA).

For a daily metric series

$$d_t = \frac{1}{N_t} \sum s_{u,k} \le \text{day } t \; g(s_{u,k}),$$

Compute its exponentially-weighted moving average (EWMA)

$$EWMA_t = \lambda d_t + (1-\lambda)\, EWMA_{t-1}, \quad 0 < \lambda \le 1,$$

and flag an anomaly when

$$|d_t - EWMA_t| > z_{0.995}\sqrt{\frac{\lambda}{2-\lambda}}\,\hat{\sigma},$$

where $\hat{\sigma}$ is the in-control standard deviation estimate and $z_{0.995} \approx 2.576$ the two-sided 99% shewhart limit.

Assume daily average CTR values:

$d_1 = 0.5$, $d_2 = 0.6$, $d_3 = 1.2$ (sudden spike)

EWMA formula:

$$EWMA_t = \lambda d_t + (1-\lambda)\,EWMA_{t-1}$$

Let $\lambda = 0.3$, $EWMA_1 = 0.5$.

Then:

$$EWMA_2 = 0.3\,(0.6) + 0.7\,(0.5) = 0.18 + 0.35 = 0.53$$

$$EWMA_3 = 0.3\,(1.2) + 0.7\,(0.53) = 0.36 + 0.371 = 0.731$$

Let $\hat{\sigma} = 0.1$. Threshold:

$$T = 2.576 \cdot \sqrt{\frac{\lambda}{2-\lambda}} \quad \therefore \hat{\sigma} = 2.576 \cdot \sqrt{\frac{0.3}{1.7}}$$

$$0.1 \approx 0.108$$

Check

$$|d_3 - EWMA_3| = |1.2 - 0.731| = 0.469 > 0.108 \Rightarrow \text{Anomaly}\downarrow.$$

☑ CTR on Day 3 is statistically unusual.

(5) Links to other complementary techniques.

Overlap	Why Combine?
Human Evaluation	Logs spot what content users flagged; human judge quality of that content.
UGR / Focus Groups	Logs reveal a drop-off step; qualitative work uncovers why users quit there.
External Data	Logs isolate the slowest 0.1% latency; external analysis focuses remediation.

(6) One-sentence takeaways

Logs-based analysis turns raw events streams into statistically rigorous metrics - fast, cheap, and at scale - making it the engineer's microscope that pinpoints where, when and for whom an experimental effect (or bug) actually occurs.

(3) Human Evaluations.

(1) What it is Human Evaluations?

-> Human evaluation means asking real people - judges, crowd workers, domain experts, or in-house graders - to score the quality of an experience (search results, recommendations, UI screenshots, generated text etc)

It delivers signals that logs cannot measure automatically, such as:

Dimension	Example question to rater
Relevance / usefulness	"Does this answer satisfy the query?"
Fluency / readability	"Is the text grammatically correct?"
Aesthetics / trust	"Would you consider this design professional?"

(2) Why you need it?

- Quality is multi-dimensional: click-through rate might rise even while relevance drops (clickbait)
- Early feedback: before shipping to millions, a panel of 50 annotators can reveal glaring issues.
- Ground truth for ML: labels from human ratings and validate models that later power live metrics.

(3) Real-world analogy backed by math.

Analogy: Think of three movie critics (raters) scoring films from 1-5 stars.

Math concept: Inter-rater agreement - we want to know if those scores are consistent.

Let each critic's score be a random variable X_{ij} for movie j and rater r. A common statistic is Cohen's κ (2 raters) or Fleiss's κ (>2).

For two raters:

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$, P_o = observed agreement, P_e = expected agreement by chance

- $\kappa = 1$: perfect agreement
- $\kappa \geq 0$: only chance agreement
- $\kappa < 0$: worse than chance.

A high κ tells you that the human-eval signal is reliable enough to act on.

Problem : Cohen's κ (2 raters).

Two raters evaluate 10 items (e.g. search results or responses) as either:

- ✓ Good (label 1)
- ✗ Bad (label 0)

We want to measure how much they agree, and whether it is better than random chance.

Step 1 : Raw Ratings

Let's say:

Item	Rater 1	Rater 2
1	1	1
2	0	0
3	1	1

Math Concept: Latent Variable Model

Let:

- X = observed behavior (e.g. time on page)
- Z = latent user state (e.g. confused or clear-headed)

GQA model:

$$P(X) = \sum_{z \in Z} P(X \mid Z = z) \cdot P(Z = z)$$

This is a mixture model, where total behavior is a weighted sum of conditional behaviors based on hidden states.

Real-World Analogy

Suppose:

- 60% of users are clear-headed → spend ~ 40 seconds on task
- 40% are confused → spend ~ 70 seconds

Then:

$$E[X] = 0.6 \cdot 40 + 0.4 \cdot 70 = 24 + 28 = 50$$

You log 52 seconds average in your A/B test — seems fine. But UGR might reveal the 40% confused users are struggling — something A/B doesn't show.

(3) Examples

Suppose you're testing a redesign of a checkout button

- A/B test shows no change in conversion
- UGR interviews reveal: "I didn't even see the new button — it looks like an ad!"

So you find a UI visibility problem A/B tests couldn't detect.

(4) How it connects with other techniques.

- Logs-based analysis: shows behavior data → UBR explains the human reason behind it.
- Human evaluation: rates perceived quality → UBR gives deep context and emotions.

(5) Key Concepts

- Qualitative insight: Interviews, eye tracking, think-aloud ~~aloud~~ sessions.
- Small sample sizes: Usually 5-20 participants
- Depth over breadth: Focus on why not just what
- Triangulations: Combine with logs and metrics for full picture.

(6) Final Summary

A/B Testing	User Experience Research (UER)
Quantitative	Qualitative
large scale	small sample, deep insight
Show what happened	Explains why it happened
works with logs and metrics	works with interviews, observation, emotion.

(5) Focus group 132

(1) What are Focus Groups?

A Focus Group is a small group of users (usually 5-10) gathered to discuss and give feedback on a product or feature.

They are led by a moderator who asks:

- What do you think of this feature?
- Was anything confusing?
- How does this compare to other tools you've used?

Why Focus Groups?

- A/B tests tell what user did.
- Focus groups reveal what users think and feel — and why.

They allow:

- open-ended discussion
- Observation of group dynamics
- Exploration of new ideas, pain points, misunderstanding.

Real-World Analogy

Think of a new product as a movie trailer.

- A/B testing is like measuring how many watched the trailer.
- Focus groups are like recording their live reaction and asking:
  - "What did you like?"
  - "What confused you?"
  - "Would you watch the full movie?"

⇒ You get deep qualitative insights that data alone can't provide.

(2) Math Concept Behind Focus Groups.

The main math concept applied in focus groups is Exploratory Data Analysis (EDA) and Latent variable Discovery.

What's happening Mathematically?

In focus groups, you're not testing a specific hypothesis. You're exploring the unknown space of possible variables — especially those that are unobserved or qualitative like:

- Trust
- Confusion
- Perceived value
- Emotional response

Math Model (Latent Variables)

You observe behavior X, but suspect it's influenced by hidden factors Z.

$$P(X) = \sum_{z} P(X \mid Z = z) \cdot P(Z = z)$$

You don't know what z is yet — focus group discussion reveals it.

Math Example

Suppose click-through rate (CTR) dropped. Logs show:

$$P(\text{Click} \mid \text{Button Seen}) = 0.4$$

But you don't know why it is low. So you model

$$P(\text{click}) = f(\text{Trust}, \text{Clarity}, \text{Need})$$

You run a focus group, and people say:

> "We didn't trust the new button color."

Now you define a new variable Z = Trust Level.

You can now build future A/B tests that measures and test trust levels.

Final Analogy

Just like in math:

| You explore a function's shape before optimizing it...

In UX:

| You use focus groups to explore user reasoning before running large-scale experiments.

(3) How it Links with other Techniques.

Technique		What it Reveals.
A/B testing	→	User behavior (clicks, time etc)
Logs Analysis	→	Detailed interaction data
UER	→	One-on-one task flow understanding.
Focus Groups	→	Group discussion, emotion / trust.
Surveys	→	Structured opinions across many.

dhawa doing rising

5) Surveys

(1) What are surveys (in Experimentation)?

-> Surveys are structured tools (like forms or questionaires) that ask users for explicit feedback about their experience.

They are used to gather:

- Perceptions (e.g. was the feature helpful?)
- Satisfaction (e.g. Rate from 1 to 5)
- Intentions (e.g. would you recommend us?)
- Qualitative comments.

Why are surveys important?

A/B tests show what users do, but surveys reveal what users think or feel.

They help:

- Validate or explain experiment outcomes.
- Measure latent variables (e.g. satisfaction, clarity, trust)
- Build subjective metrics like Net Promoter score (NPS)

Real-world Analogy

Imagine changing a feature on your website.

- A/B test shows: click rate dropped by 2%.
- But you don't know why.

You send a survey, and users say:

> "It was hard to find the new button."
> "The label was confusing."

That's qualitative insight from structured feedback.

(2) Math Concept in Surveys

Surveys touch on several math/statistics areas:

Concept	Role in survey
Ordinal scales	→ Likert-type response (e.g. 1 to 5 ratings)
means/Variance	→ Average satisfaction, consistency of scores.
Sampling bias	→ Who answers affect results.
Missing data	→ Not all users respond.
Correlation	→ Survey score ↔ behavior (e.g., High rating ↔ High retention)
Latent variables	→ Surveys often measure unobservable constructs like "trust".

(3) Math Example: Satisfaction vs. Clicks

Suppose you ask 100 users:

- "How satisfied are you? (1 to 5)"
- And also measure: "Did they click the new feature?".

Now, you can correlate:

$$r = \text{corr}(\text{satisfaction}, \text{click})$$

Or run regression

$$P(\text{click}) = \alpha + \beta \cdot \text{satisfaction}$$

Extramal Data

(1) What is Extramal Data?

Extramal data means looking at the outliers - users who performed extremely well or poorly during an experiment.

Instead of focusing on the average user, we analyse:

- Top 1% most engaged users.
- Bottom 1% users who bounced quickly
- User who experienced failures/crashes.

Why use Extramal Data?

- surface hidden bugs or pain points that only a few users experience.
- Detect edge cases that affect business risk
- Identify super-users or at-risk users.

These insights often don't show up in the average metrics used in A/B testing.

Real-world Analogy

Let's say you redesign a ride-sharing app.

- On average, time-to-book improved.
- But extramal data reveals
  - ↳ Some users waited 5x longer in rural areas.
  - ↳ Others completed rides faster than ever.

Without analysing the tails, you'd miss both.

② Math Concept: Tail Analysis / Quantiles.

Extremal data is about studying distribution tails, not just the mean.

Example:
Let X be a performance metric (e.g. load time)
we study:
- Lower tail: $P(X < x)$, like 1st percentile
- Upper tail: $P(X > x)$, like 99th percentile

use quantiles:

$$P_1 = Q_{0.01}(X), \quad P_{99} = Q_{0.99}(X)$$

③ Interlinked Concepts

Related Topic	Connection to Extremal Data
Logs-based Analysis	→ Needed to trace what happened during outlier sessions
Human Evaluation	→ Helps explain why the outliers behaved differently
Survey	→ Can targets only top/bottom 5% to collect qualitative data.
UBR	→ Helps dig into specific user journeys of outliers

Summary

→ This section emphasizes that no single method is enough to understand user behavior or make reliable product decisions.

- A/B tests are powerful, but incomplete.
- Complementary techniques like logs analysis, surveys, UERP, focus groups, external data — all add different perspectives.

Core Message:

The best insights come when we combine all complementary techniques with experiments to explain not just what happened but also why it happened.

Real-world Analogy

Imagine you're a doctor diagnosing a patient:

Tool	Purpose
Blood test (A/B test)	Objective measurement
Interview (focus group)	Understand symptoms and experience.
MRI (Logs)	Look inside how things function
Survey (Questionnaire)	Quantify pain, satisfaction, mental state.
History (External data)	Look at rare but critical past conditions.

No single test gives the full picture — diagnosis comes from putting it all together.

(2) Mathmatical concept applied in "Putting it all together",

(1) Ensemble learning (from ML) ⇒ Most Applied.

Concept: Combine multiple weak or partial predictors into one strong predictor,

Math form:

$$y = \sum_{i=1}^{n} w_i \cdot x_i$$

where x_i is each technique (logs, survey, etc) and w_i is a weight.

Used when: Aggregating signals (e.g. A/B test + survey + logs) to estimate outcome or satisfaction.

(2) Vector Spaces & Orthogonality

Concept: Treat each source (logs, survey, etc) as a vector in a feature space. If two are orthogonal, they are independent.

Math:

$$\vec{m_i} \cdot \vec{m_j} = 0 \Rightarrow \text{independent insights}$$

Used when: Ensuring that each method add's new, non-overlapping information.

(3) Principal Component Analysis (PCA) ⇒ Heavily used in Analysis

Concept: Reduce dimensionality by combining overlapping techniques into orthogonal axes that explain most variance.

Math:

Decompose data matrix X into eigenvectors of the covariance matrix

$$X = U \Sigma V^T$$

Used when: Visualizing and simplifying many metrics into interpretable components (e.g. usability, trust, performance)

(4) Bayesian Inference

Concept: Use prior belief (from surveys, expert judgement etc) and update with evidence (e.g. A/B test result).

Math:

$$P(H/D) = \frac{P(D/H) \cdot P(H)}{P(D)}$$

Used when: Merging qualitative and quantitative signals - e.g. starting with expert suspicion and confirming with log data.

(5) Multi-Model Data Fusion

Concept: Combine different data types (numerical from logs, categorical from surveys, text from focus groups).

- Numerical fusion: Normalize and add
- Latent model fusion: Embed all types in a shared space.

(6) Co-variance and Redundancy Analysis.

Concept: Measure how much overlap (redundancy) exits between signals.

Math:

$$Cov(X_i, X_j) = \Sigma [(X_i - \mu_i)(X_j - \mu_j)]$$

used when: Deciding whether to include both logs and click data (if they're too correlated, drop one).

7) Decision Theory & Utility Functions

Concept: Combine multiple uncertain outcomes into a single decision with cost-benefit trade-offs.

Math:

$$\text{Utility} = \sum_{i} p_i \cdot u_i$$

where p_i is the probability of outcome i, and u_i is the utility

Used when: Balancing metrics — e.g. clicks vs. satisfaction vs. revenue.

Examples

Ensemble Modeling & Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

where:

- y = estimated satisfaction score
- β_0 = intercept (baseline)
- $\beta_1, \beta_2, \beta_3$ = weights (learned from past data)

~~Let's~~ assume

$\beta_0 = 1.0$, $\beta_1 = 1.2$, $\beta_2 = 0.05$, $\beta_3 = 0.8$

Real analogy:

Each x_i is noisy and biased on its own.

But together:

- A/B test says "They converted"
- Logs say "They stayed 50 seconds"
- Survey says "Rated 4 stars".

Now you can predict y more accurately than with any single method.

Sim • A user clicked → $x_1 = 1$

• Spent 40 seconds on the page → $x_2 = 40$

• Gave a rating of 4 → $x_3 = 4$

And assume model:

$$y = 0.5 + 0.3 \cdot x_1 + 0.02 \cdot x_2 + 0.4 \cdot x_3$$

Then:

$$y = 0.5 + 0.3(1) + 0.02(40) + 0.4(4) = 0.5 + 0.3 + 0.8 + 1.6 = 3.2$$

So we estimate the user's true satisfaction is around 3.2 out of 5.

③ Interlocking Concepts

Technique	what it Adds to A/B Test
Logs	→ Deep behavior tracking
UBR	→ Discover hidden usability blockers
Focus Groups	→ Reveal emotional response & group trends
Survey	→ Quantify subjective variables like trust
External data	→ Uncover critical bug / best user experience.